

Selection on Observables I: The Assumption

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The question: Can we recover causal effects anyway, by adjusting for observed covariates X ?

The First Commandment of Causal Inference

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All of these are *estimation* choices. They serve a causal claim only if the underlying *identification assumption* is credible.

Setting

- Units: $i = 1, \dots, N$
- Treatment: $D_i \in \{0, 1\}$, **not** randomly assigned
- Potential outcomes: Y_{0i}, Y_{1i}
- Pre-treatment covariates: X_i (a vector)
- Quantity of interest: $\tau_{ATE} = \mathbb{E}[Y_{1i} - Y_{0i}]$

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A covariate X is a **confounder** if it is associated with both treatment D and the potential outcomes Y_d .

The Assumption: Conditional Ignorability

Identification Assumption (Conditional Ignorability / Selection on Observables)

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This assumption has many names:

- Conditional ignorability (CI)
- Selection on observables (SOO)
- No unmeasured confounders
- Conditional exchangeability
- “Backdoor criterion satisfied” (in DAG language)

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Read it as: *among units with the same X , treatment assignment is as good as random.*

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By design: a conditionally randomized experiment (e.g., block-randomized with different treatment probabilities per stratum). Then CI holds by construction.

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- CI is **not verifiable from the data**. No amount of checking the observed covariates can rule out unobserved confounders.
- The credibility of CI depends on the *story*: what determined treatment, and have you measured all of it?

Common Support

CI alone is not enough. We also need:

Identification Assumption (Common Support / Overlap)

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This can pose trouble when the probability of treatment (or control) is very high or very low for some units — there may be little or no information about the missing counterfactual in that region of X .

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$$\begin{aligned} \tau_{ATE} = \mathbb{E}[Y_{1i} - Y_{0i}] &= \sum_x \mathbb{E}[Y_{1i} - Y_{0i} \mid X=x] \Pr(X=x) \\ &= \sum_x (\mathbb{E}[Y_i \mid D=1, X=x] - \mathbb{E}[Y_i \mid D=0, X=x]) \Pr(X=x) \end{aligned}$$

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Within-stratum effects, averaged over the population distribution of X . The RHS is all observable (by common support), so the ATE is identified.

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This is *collider bias* / *post-treatment conditioning*. DAGs make this transparent.

Estimation Under CI: The Idea

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Subclassification (stratification) does this literally when X is discrete:

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This is transparent and requires no modeling. But it breaks down with many covariates or continuous X (curse of dimensionality). That's where matching, weighting, and regression come in — next time.

Example: Smoking and Mortality (Cochran 1968)

Unadjusted death rates per 1,000 person-years:

	Canada	U.K.	U.S.
Non-smokers	20.2	11.3	13.5
Cigarette only	20.5	14.1	13.5
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Or could something else explain this?

The Confounder: Age

Mean ages by smoking group:

	Canada	U.K.	U.S.
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Age is associated with both smoking type and death rate — it is a confounder.
Can we adjust for it?

Subclassification Reveals the Effect

Fabricated data inspired by the U.S. figures (death rates per 1,000 person-years):

Age group	Rate (Cig)	Rate (Non)	Effect	<i>N</i> (Cig)	<i>N</i> (Non)
< 45	5	3	+2	40	10
45–54	10	5	+5	30	20
55–64	20	10	+10	20	30
65+	45	23	+22	10	40
Pooled	13.5	13.5	0	100	100

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The confounding hid an effect of nearly 10 deaths per 1,000 person-years.

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Variables we stratified on balance by construction. The informative question is balance on variables we did *not* target.

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This is the hard part — not the estimator.

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ID strategy: Selection on observables.

"Abduction was large-scale and seemingly indiscriminate; 60,000 to 80,000 youth are estimated to have been abducted. . . Youth were typically taken by roving groups of 10 to 20 rebels during night raids on rural homes. Adolescent males appear to have been the most pliable, reliable and effective forced recruits, and so were disproportionately targeted."

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That's the key – it is arguing that *conditional on age and location*, abduction “as is” random; no other confounders.

Blattman & Annan: Evaluating the SOO Claim

Data: panel survey of male youth in war-affected regions of Uganda.

- `abd`: abducted by LRA (the treatment)
- `c_ach, ..., c_pal`: location indicators
- `age`: age in years
- `fthr_ed, mthr_ed`: parental education
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- consider a given village and age group
- for youth in that group, compare the mean outcome for the abducted to the non-abducted. That gives you an estimated effect in one stratum of location $\times$ age
- then average these strata-wise effects across all location–age groups, weighted by their share of the sample:

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$$\hat{\tau}_{ATE} = \sum_{j=1}^J \frac{N_j}{N} (\bar{Y}_{1j} - \bar{Y}_{0j})$$

where j indexes location–age strata.

Blattman & Annan: “Adjusting”, concretely

To be clear, “adjusting for location and age” means,

- consider a given village and age group
- for youth in that group, compare the mean outcome for the abducted to the non-abducted. That gives you an estimated effect in one stratum of location $\times$ age
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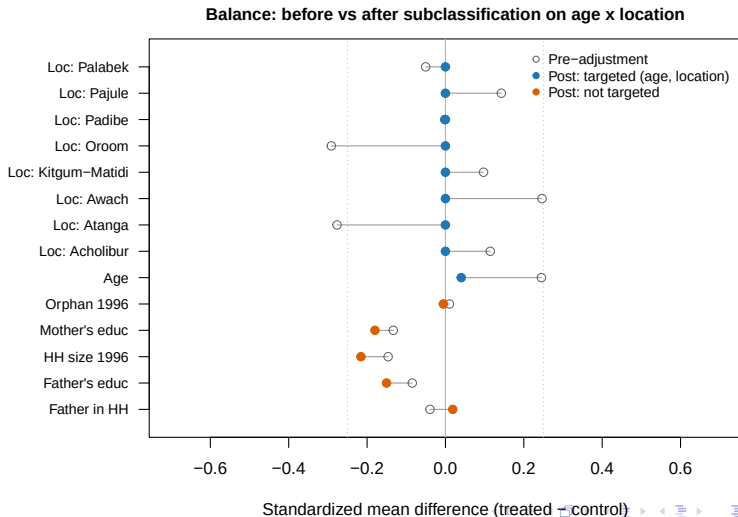
where j indexes location–age strata.

Subclassification estimates (8 locations by 4 age bins, $N = 741$):

- Effect on years of education: $\hat{\tau} \approx -0.84$ (abduction reduces schooling)
- Effect on emotional distress index: $\hat{\tau} \approx +0.50$ (abduction increases distress)

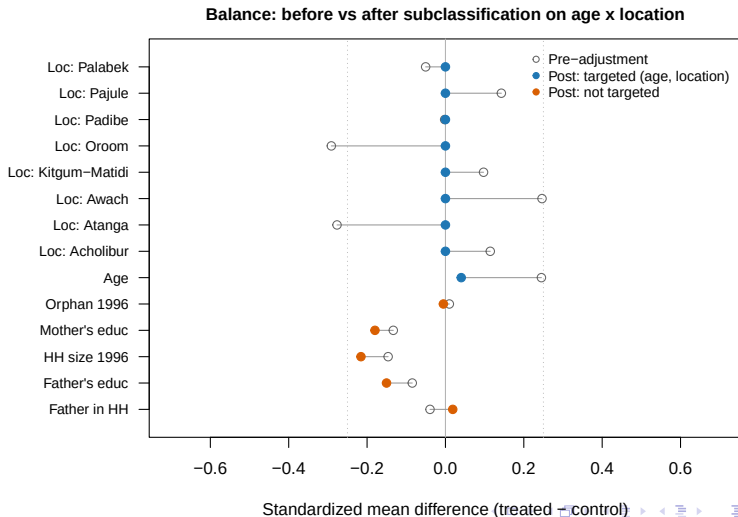
Blattman & Annan: Checking Balance

Apply the “same adjustment on X ” procedure to every pre-treatment variable, before vs. after subclassifying on age $\times$ location:



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Targeted variables (age, location): post-adjustment SMDs at or very near zero.
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- Orphan status and father-in-HH: close to balanced before *and* after. Consistent with the SOO story.
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Falsification, not a test. Poor balance on (untargeted) observables can warn you there is a problem; cannot tell you there is no problem.

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Contrast with: “We ran a regression with 50 controls and the coefficient was significant.” That is not an argument for SOO or any other ID strategy.

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- **Don't condition on post-treatment variables.**
- **Once you believe CI**, the estimation question (matching, weighting, regression) is secondary. We turn to that next.
- **If you don't fully believe CI** — and you shouldn't — do something else or use sensitivity analysis. We'll get there.